AMAZON SALES PERFORMANCE AND CUSTOMER BEHAVIOUR ANALYSIS

A Data Driven Study Using SQL & Power BI

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- Performance Data Insights
- Data Preparation & transformation
- Amazon Business Performance Overview
- Revenue Distribution Across Location
- Revenue Trend by Category (Yearly Comparison)
- Monthly Revenue Performance
- Key Drivers of Customer Dissatisfaction

AGENDA

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- Categories Growth Rate (Year-Over-Year)
- Delivery Efficiency Across Product Categories
- Pricing VS Revenue Performance by Category
- Individual Product Category Performance Analysis
- Detailed Product Performance
- Top Performing Products by Revenue
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ABOUT AMAZON

Amazon is a global e-commerce and technology company focused on online retail, cloud computing, digital streaming, and logistics. Founded in 1994, it has grown into one of the world's largest marketplaces, offering a wide range of products across multiple categories. Amazon provides a seamless shopping experience through fast delivery, competitive pricing, and customer-centric services.

With innovations in supply chain management, data-driven personalization, and platforms like Amazon Web Services (AWS), it continues to transform how businesses operate and how customers shop, making it a leader in the global digital economy.

BUSINESS CHALLENGE

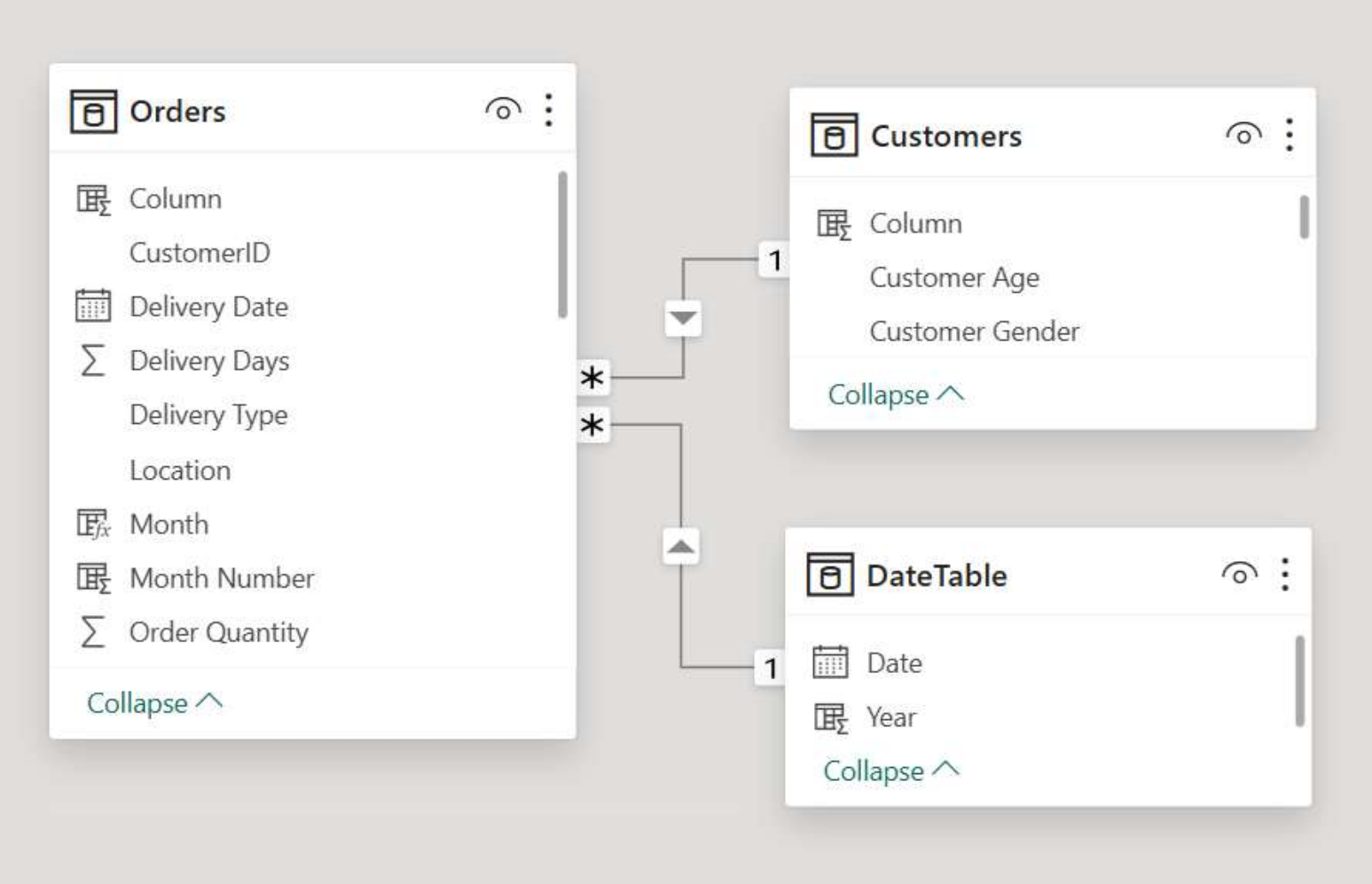
The objective is to identify key revenue drivers, customer behavior patterns, and operational inefficiencies to improve overall business performance and customer satisfaction. Stakeholders require a structured report with three tabs. The **Main Tab** provides an overview of total sales, revenue, and customer satisfaction, with filters for product category and date.

The **Product Category Tab** enables analysis of category-wise performance, customer demand, and areas of improvement using filters such as gender, delivery type, category, and age group.

The **Individual Product Tab** focuses on detailed product-level insights, including revenue, ratings, category, and purchase behavior to support targeted decision making.

The report is expected to be well-structured, visually clear, and tailored to different stakeholders for effective analysis.

DATABASE SCHEMA



The data model follows a **star schema**, with **Orders as the fact table** and **Customers and Date as dimension tables**.

The **Date table was derived from the order date to extract Year and Month**, enabling time-based analysis. This structure ensures accurate filtering and efficient insights across the dashboard.

PERFORMANCE DATA INSIGHTS

Data Overview

Dataset Scope: 2015 - 2020 sales across multiple categories

Revenue & Scale: 107.24M revenue/ 113K orders & customers

Top Drivers: Health & Beauty and Fashion lead revenue

Growth Insight: Electronics shows highest YoY growth (47%)

Customer Satisfaction: Average rating is **2.73**

Delivery: Average delivery time is **9.5 days** (up to 15 days abroad)

Quality Issues: Returns driven by **defective/missing items (5.8K - 6.1K)**

Key Insight: Revenue is concentrated in select categories, while delivery delays and product quality issues drive customer dissatisfaction.

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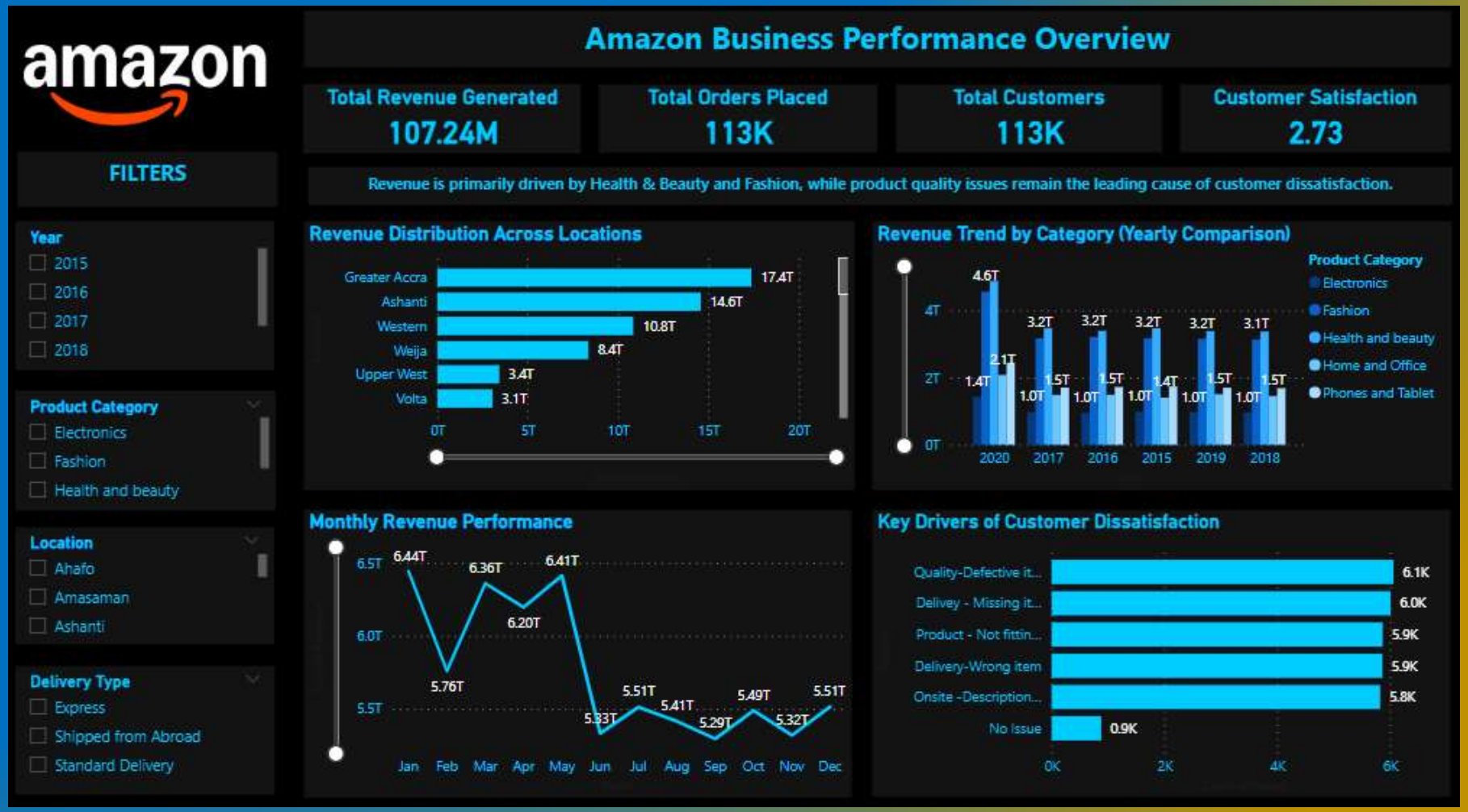
DATA PREPARATION TRANSFORMATION

The dataset was cleaned by handling missing values and inconsistencies, and structured into a data model using fact and dimension tables. Key features such as *Delivery Days, Year, and Month* were derived, and DAX measures were created for KPIs like *Total Revenue, Orders, Customers,* and *Average Rating*. Time intelligence was applied for Year-over-Year analysis, with relationships built across tables for accurate insights.

The report was developed as a **three-tab dashboard**. The **Main Tab (Amazon Business Performance Overview)** provides an overview of revenue, sales, and customer satisfaction with trend and distribution analysis. The **Product Category Tab (Product Category Performance Analysis)** focuses on category-level performance, including demand, growth, delivery efficiency, and pricing insights. The **Individual Product Tab (Individual Product Performance Analysis)** enables detailed product-level analysis using tables, top-performing products, and satisfaction vs revenue segmentation.

AMAZON BUSINESS PERFORMANCE OVERVIEW



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AMAZON BUSINESS PERFORMANCE OVERVIEW

This dashboard provides a comprehensive view of business performance using key metrics, filters, and visual analysis. The KPIs highlight overall business health, showing **Total Revenue of 107.24M**, **113K orders and customers**, and an average **customer satisfaction of 2.73**, giving a quick snapshot of performance.

Slicers such as **Year, Product Category, Location, and Delivery Type** enable dynamic filtering, allowing users to analyze trends over time, compare category performance, assess regional contribution, and evaluate operational impact.

The visualizations further support analysis by showing **revenue distribution across locations**, **category wise trends over time**, **monthly revenue patterns**, and **key drivers of customer dissatisfaction**, providing a complete understanding of revenue performance, trends, and customer experience.

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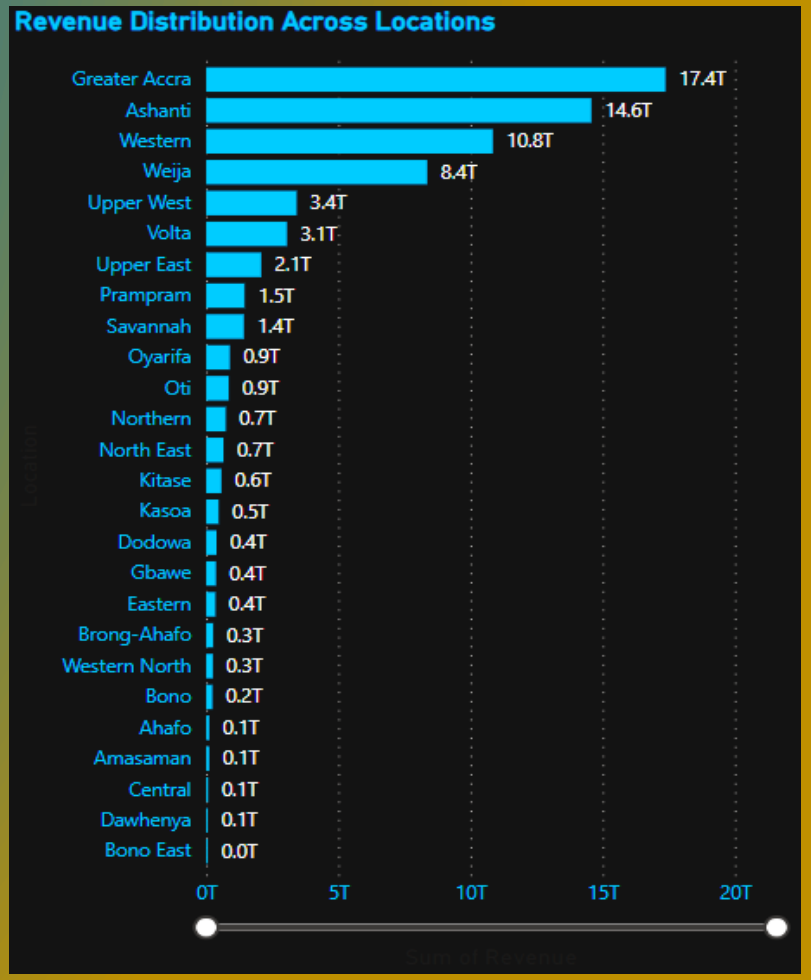
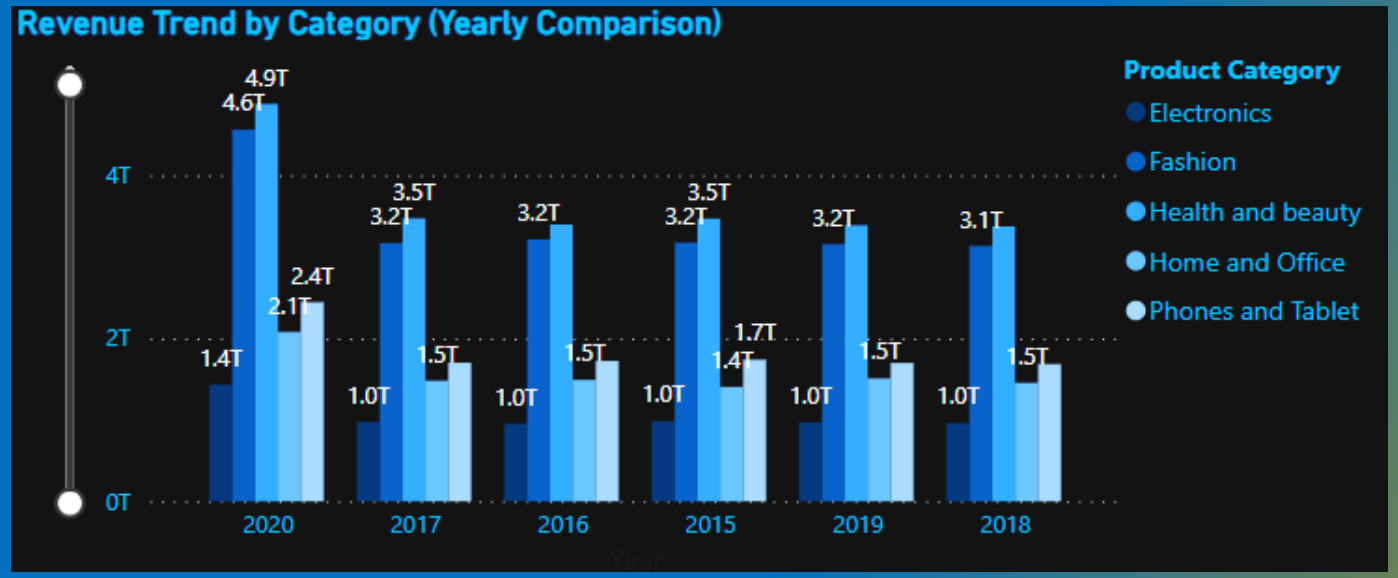
REVENUE DISTIBUTION ACROSS LOCATION

- **High concentration** at Greater Accra (17.4T) & Ashanti (14.6T) contribute **50%+ of total revenue.**
- Adding Western (10.8T) and Weija (8.4T) brings contribution to **75%+**, showing **limited regional diversification.**
- Revenue drops from **17.4T to less than 3.5T** beyond top regions, highlighting **uneven market penetration.**
- Majority of locations generate **less than 1T**, with several near **0.1T**, indicating **underutilized markets.**
- Mid-tier regions (Western, Weija) offer **scalable expansion potential** compared to already saturated top regions.
- Revenue is concentrated in a few regions with a **clear long tail distribution, indicating dependency risk and significant untapped growth potential across smaller markets.**



REVENUE TREND BY CATEGORY (YEARLY COMPARISON)



- Health & Beauty (4.9T) and Fashion (4.6T) dominate revenue in recent years.
- Electronics and Home & Office remain consistent at 3.1T - 3.5T, showing limited growth.
- Phones & Tablet stays lowest at 1.4T - 1.7T, indicating weak category performance.
- Overall category revenues show gradual increase toward 2020, with no sharp volatility.
- Top categories generate 2 - 3x higher revenue than lowest category.

Revenue is driven by lifestyle categories (Health & Beauty, Fashion), while other categories show stable but limited growth, indicating scope for repositioning and optimization.

MONTHLY REVENUE PERFORMANCE

- January (**6.5T**) and May (**6.4T**) show highest revenue levels
- Revenue drops significantly to **5.3T in June** (18 - 20% drop from peak).
- Post June, revenue stabilizes between **5.3T and 5.5T.**
- February (**5.7T**) and September (**5.29T**) indicate weaker demand periods.
- Overall variation of **1.2T** between peak and lowest months.

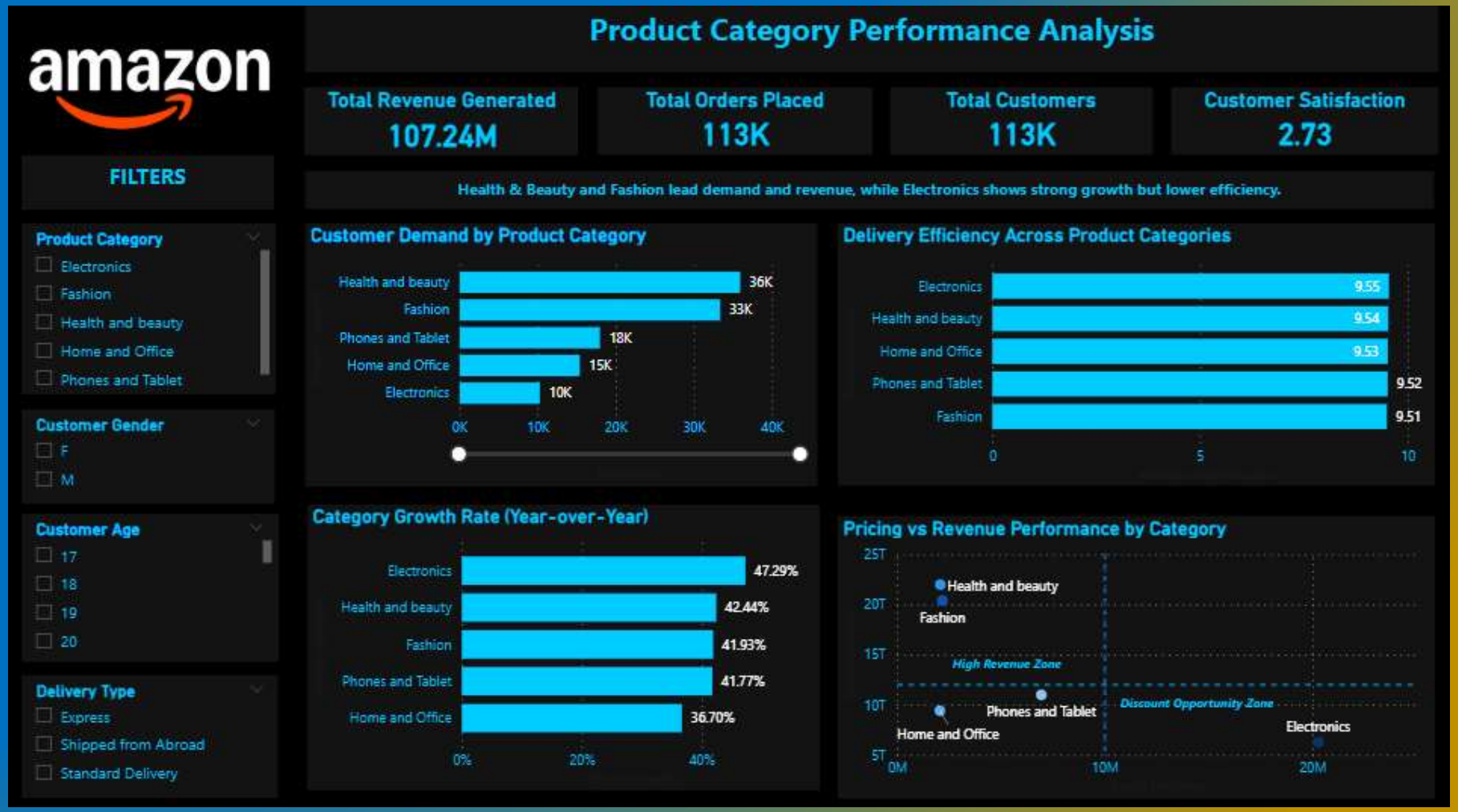
Revenue shows clear seasonality with strong early year performance and a mid-year drop, indicating opportunities for targeted promotions and demand stabilization strategies.

KEY DRIVERS OF CUSTOMER DISSATISFACTION

- Defective items (6.1K) and missing items (6.0K) are the leading causes.
- Wrong item and product mismatch contribute 5.8K - 5.9K cases each.
- All major issues are clustered around 5.8K - 6.1K, indicating systemic problems.
- “No issue” cases are significantly lower (0.9K), confirming genuine dissatisfaction.
- Quality related issues alone account for a majority of total complaints.

Customer dissatisfaction is primarily driven by product quality and fulfillment errors, indicating systemic operational gaps rather than isolated issues.

PRODUCT CATEGORY PERFORMANCE ANALYSIS



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PRODUCT CATEGORY PERFORMANCE ANALYSIS

This report evaluates category performance across demand, growth, operations, and pricing to provide deeper insights into business performance. The KPIs remain consistent (**Total Revenue, Orders, Customers, and Customer Satisfaction**) to ensure a **standardized view of performance across all tabs**, enabling easy comparison and continuity in analysis.

Slicers such as **Product Category, Gender, Age Group, and Delivery Type** allow dynamic segmentation, helping identify variations in customer behavior and operational impact. The visualizations focus on **customer demand, delivery efficiency, YoY growth, and pricing vs revenue**, providing a comprehensive understanding of category performance.

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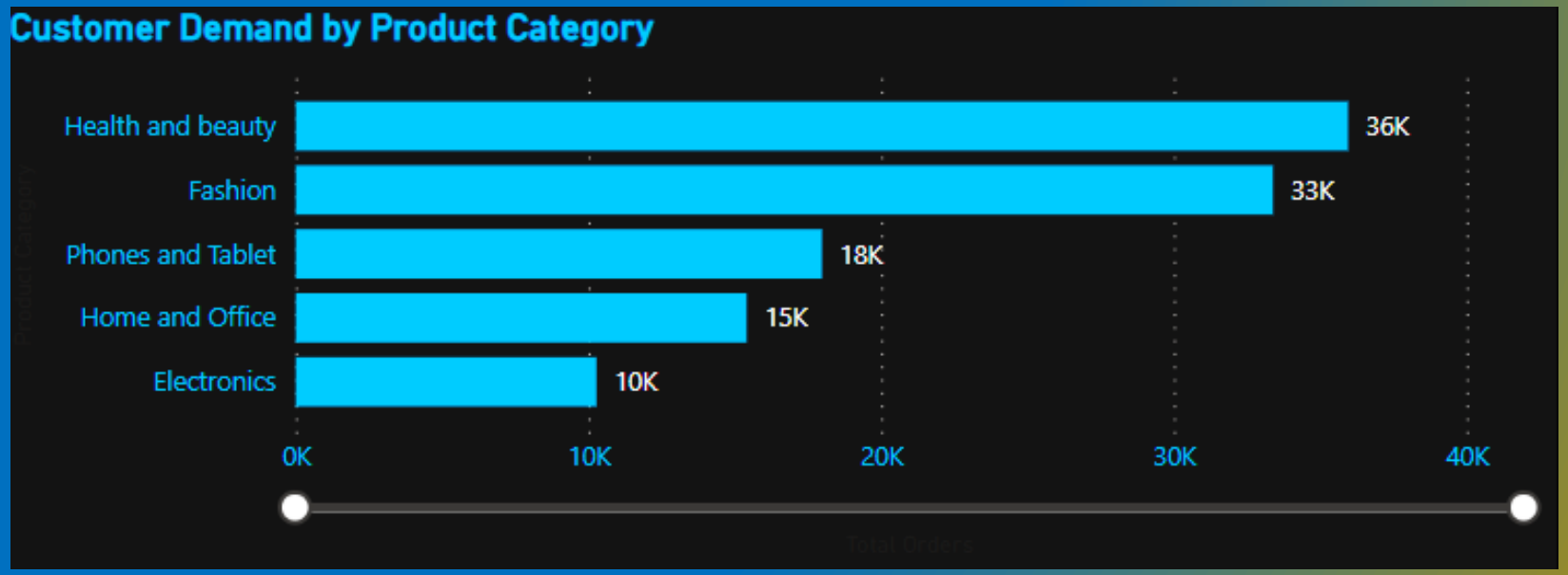
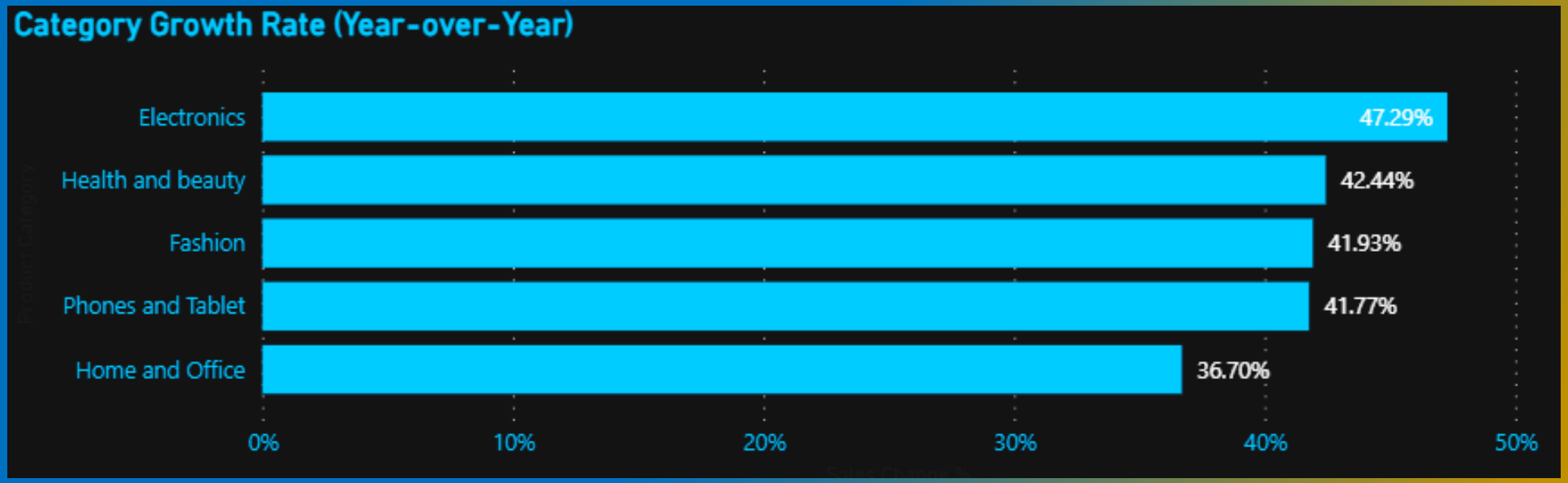
CUSTOMER DEMAND BY PRODUCT CATEGORY



- Health & Beauty (36K orders) and Fashion (33K) together contribute 55%+ of total demand.
- Phones & Tablet (18K) and Home & Office (15K) show moderate demand (2x lower than top categories).
- Electronics at 10K orders, indicating weak customer preference.
- Top category demand is 3.5x higher than Electronics, showing strong category imbalance.

Customer demand is heavily concentrated in lifestyle categories, while Electronics underperforms, indicating potential gaps in pricing, positioning, or product-market fit.

CATEGORIES GROWTH RATE (YEAR-OVER-YEAR)



- Electronics leads with 47.3% YoY growth, highest among all categories.
- Health & Beauty (42.4%) and Fashion (41.9%) show consistent high growth.
- Phones & Tablet at 41.8% indicates steady expansion.
- Home & Office at 36.7%, lagging behind other categories.
- Most categories fall within 36 - 47%, indicating overall market expansion.

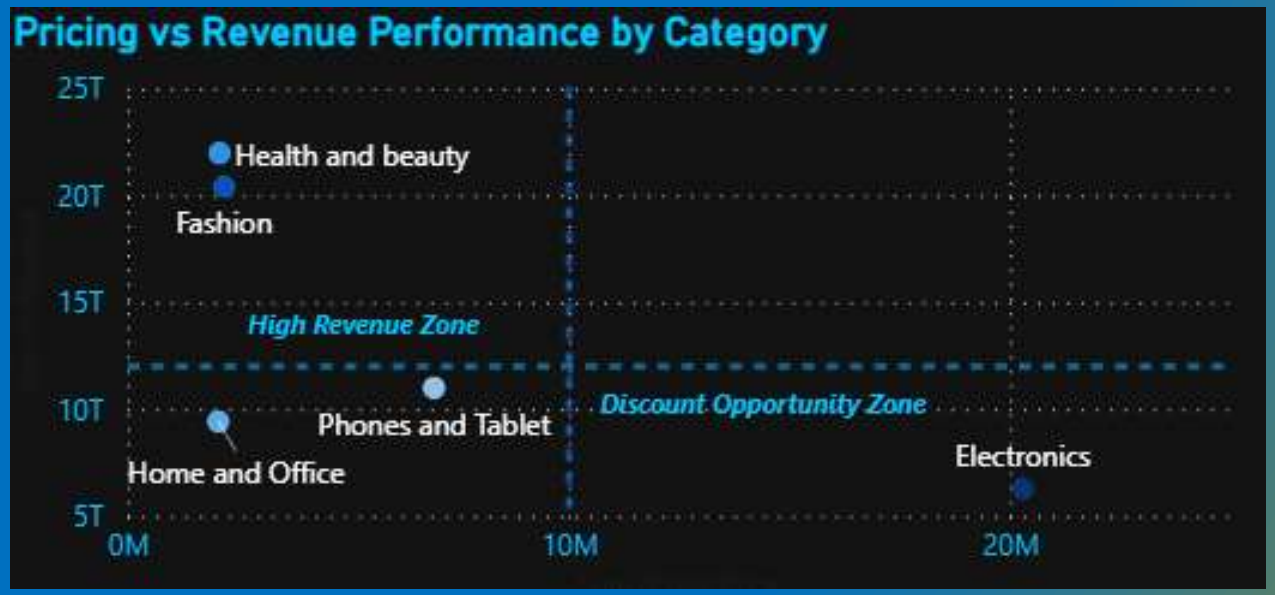
While Electronics shows the fastest growth, high-performing categories like Health & Beauty and Fashion combine strong growth with high revenue, making them more strategically valuable.

DELIVERY EFFICIENCY ACROSS PRODUCT CATEGORIES

- Delivery time is tightly clustered between **9.51 - 9.55 days** across all categories.
- Difference between fastest and slowest is **less than 0.05 days**, **indicating uniform logistics performance.**
- All categories experience **similar delivery timelines**, regardless of demand or product type.

Delivery performance is consistent across categories, indicating that customer dissatisfaction is driven more by product quality than logistics delays.

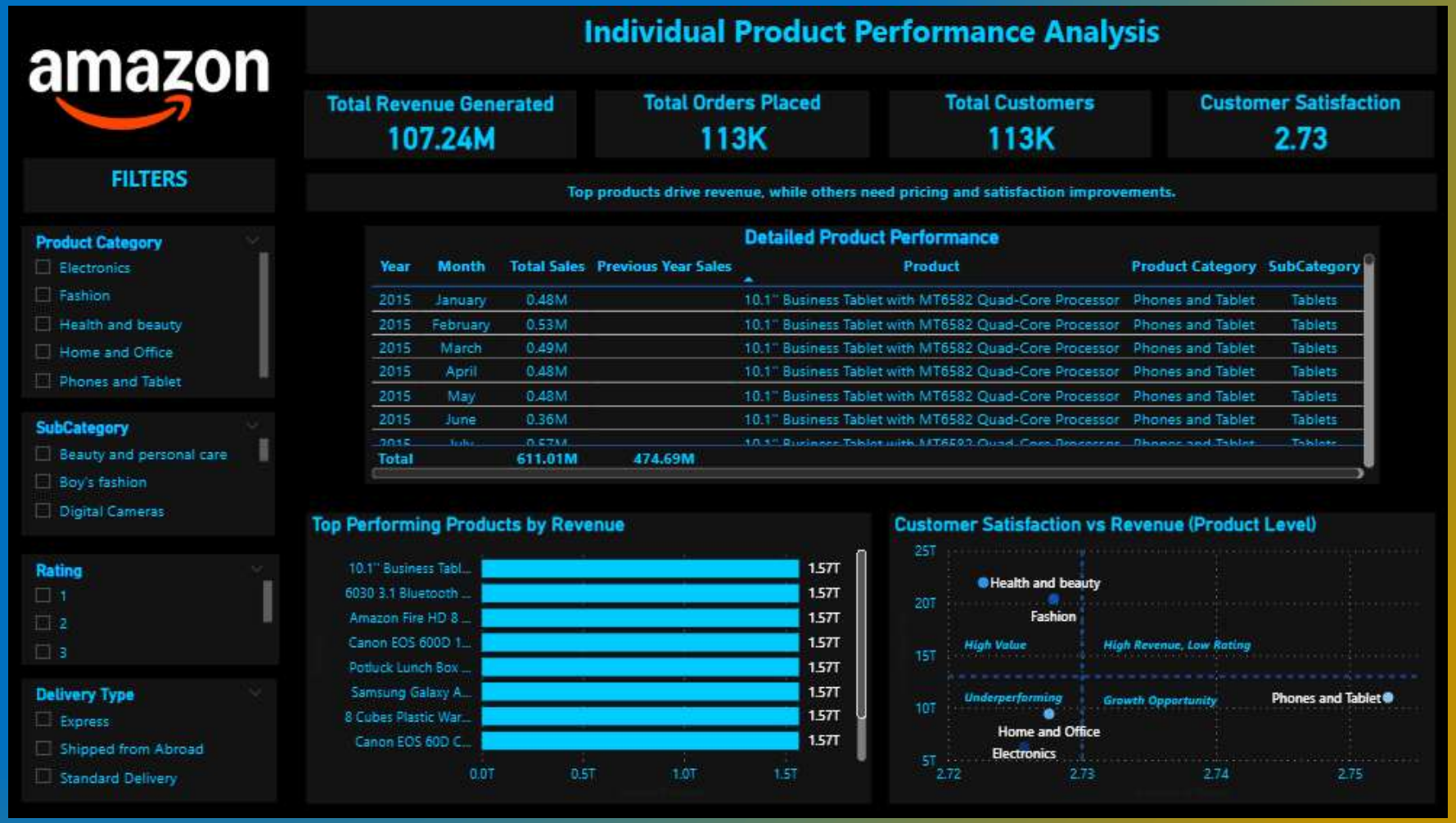
CATEGORIES GROWTH RATE (YEAR-OVER-YEAR)



- Health & Beauty and Fashion generate **20T - 22T revenue at lower price levels**, **indicating strong pricing efficiency.**
- Electronics has the **highest price (20M)** but generates only **6 - 7T revenue**, showing poor price to revenue conversion.
- Phones & Tablet (10M price, 11T revenue) performs moderately well.
- Home & Office has **low price (3 - 4M)** but also low revenue (9- 10T), limiting impact.
- Higher pricing does not translate to higher revenue across categories.

Categories with optimized pricing (Health & Beauty, Fashion) outperform high-priced categories like Electronics, highlighting pricing inefficiency and the need for repositioning.

INDIVIDUAL PRODUCT CATEGORY PERFORMANCE ANALYSIS



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INDIVIDUAL PRODUCT CATEGORY PERFORMANCE ANALYSIS

This section enables deep product-level evaluation by combining detailed visuals with dynamic filters for targeted analysis. The KPIs remain consistent (**Total Revenue, Orders, Customers, and Customer Satisfaction**) to maintain a **standardized performance baseline across all tabs**, ensuring continuity and easy comparison from overall to category to product-level insights.

The analysis includes a **detailed product table** for granular sales and YoY tracking, **top performing products** to identify revenue concentration, and **customer satisfaction vs revenue** to segment products into high value, risk, and opportunity zones. Slicers such as **Category, Sub Category, Rating, and Delivery Type** enable focused exploration to isolate performance issues and opportunities across product segments.

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DETAILED PRODUCT PERFORMANCE

Detailed Product Performance

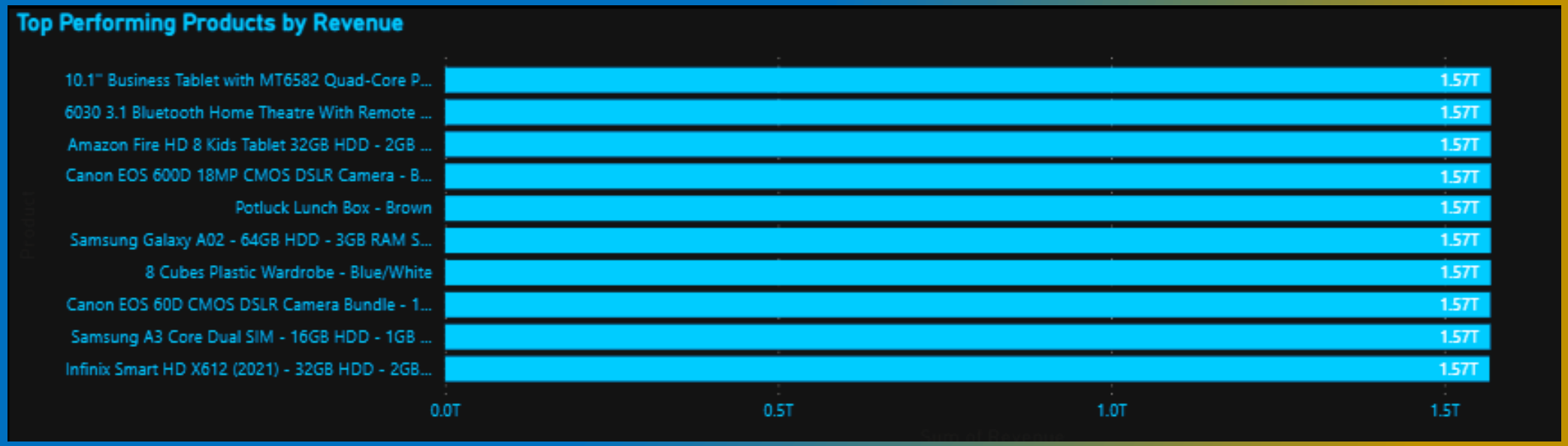
Year	Month	Total Sales	Previous Year Sales	Product	Product Category	SubCategory
2015	January	0.48M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	February	0.53M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	March	0.49M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	April	0.48M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	May	0.48M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	June	0.36M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	July	0.57M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	August	0.52M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
2015	September	0.50M		10.1" Business Tablet with MT6582 Quad-Core Processor	Phones and Tablet	Tablets
Total		**611.01M**	**474.69M**			

This table provides a **granular, product-level view** of sales and Year-over-Year performance, enabling detailed analysis beyond aggregated visuals.

- Shows exact sales values (**e.g., 0.36M - 0.57M monthly**) for individual products.
- Allows direct comparison between current and previous year sales (**611.01M vs 474.69M total**).
- Connects each product to its category and subcategory for deeper segmentation.

Acts as a drill-down layer to validate insights, analyze specific products, and support detailed decision-making after applying filters.

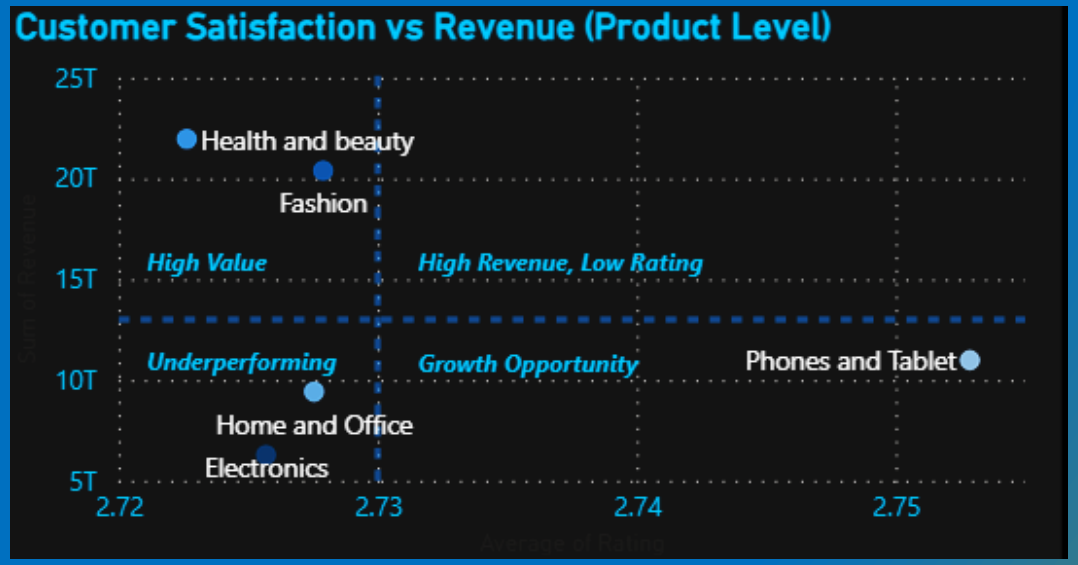
TOP PERFORMING PRODUCTS BY REVENUE



- Top products each generate **1.57T revenue**, indicating strong contribution from a limited set.
- Minimal variation across top products suggests **consistent performance within the top tier.**
- Revenue is concentrated among **few high performing stock keeping units**, showing dependency.
- Product mix spans across categories, indicating **no single category dominance at product level.**
- Lack of clear differentiation among top products may indicate **similar pricing or demand patterns**.

Revenue is concentrated among a small group of top-performing products, highlighting dependency and the need to scale similar high-performing products to drive growth.

TOP PERFORMING PRODUCTS BY REVENUE



- Health & Beauty (22T) and Fashion (20T) generate high revenue despite moderate ratings (2.72 - 2.73).
- Phones & Tablet shows **highest rating (2.75)** but moderate revenue (11T).
- Electronics (6T) and Home & Office (9T) show **low revenue with average ratings (2.72 - 2.73).**
- Higher ratings do not always translate to higher revenue.
- Products fall into High Value, Risk, Opportunity, and Underperforming zones.

High-revenue categories with lower ratings indicate quality risks, while high - rating categories with lower revenue present strong growth opportunities.

CONCLUSION

- Revenue is highly concentrated in a few categories, with **Health & Beauty (36K orders)** and **Fashion (33K)** together driving **55%+ of total demand**, making them the primary revenue engines.
- **Electronics shows the highest YoY growth (47.3%)** but generates relatively low revenue (**6 - 7T**), highlighting a clear **pricing-performance mismatch** and missed monetization opportunity.
- Mid-tier categories like **Phones & Tablet (18K orders)** and **Home & Office (15K)** show stable demand but are not fully optimized for revenue growth.
- Delivery performance is consistent across all categories (**9.51 - 9.55 days**), indicating **no major logistical bottlenecks** impacting performance.
- Customer dissatisfaction is primarily driven by **product and delivery issues (5.8K - 6.1K cases)**, suggesting **quality and fulfillment gaps rather than demand issues**.
- At the product level, revenue is concentrated among a few SKUs, with top products generating **1.57T each**, indicating **high dependency and limited product diversification**.
- Customer satisfaction remains moderate (**2.73 rating**), and analysis shows that **high revenue does not always align with high ratings**, indicating potential **long-term customer experience risks**.

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RECOMMENDATION & NEXT STEPS

- **Optimize Electronics pricing strategy** to improve revenue conversion, leveraging its strong **47% growth momentum**.
- **Invest further in Health & Beauty & Fashion**, which already drive **55%+ demand**, through promotions and inventory scaling.
- **Improve product quality and fulfillment accuracy**, targeting top issues (**6K+ cases**) to increase customer satisfaction beyond **2.73 rating**.
- **Scale high-performing products** and replicate successful SKUs (1.57T revenue each) to reduce dependency risk.
- **Leverage high-rating categories (e.g., Phones & Tablet 2.75 rating)** to drive revenue growth through better pricing and positioning.

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THANK YOU

Open to Questions